

Project: Automated Domain Risk Intelligence Pipeline

Overview

This project is an autonomous Decision Support System (DSS) designed to evaluate the legitimacy of corporate domains. Instead of relying on fragile web scraping methods, which are often blocked and difficult to maintain, this pipeline utilizes OSINT (Open Source Intelligence) techniques and AI-driven analysis to assess risk factors associated with domain registration.

The Problem

Manual background checks on companies are time-consuming and inconsistent. Traditional scraping tools often break when websites update their frontend, and they frequently violate the platform's Terms of Service. There was a need for a more stable, scalable way to verify if a company is legitimate.

The Solution: Workflow Architecture

The pipeline is built in n8n, utilizing a modular design that handles data ingestion, enrichment, analysis, and storage.

- **Ingestion Layer (Flexible Input):** The workflow accepts either a direct URL or a company name. If a URL is not provided, the pipeline automatically triggers a Google Search via **SerpApi** to identify the correct domain, ensuring a seamless user experience.
- **Enrichment Layer (RDAP Protocol):** Rather than scraping, the pipeline uses **RDAP (Registration Data Access Protocol)**. This queries authoritative registrar data to retrieve clean, reliable facts such as domain age, registration dates, and expiration data.
- **Intelligence Layer (AI-Driven Analysis):** The raw technical data is passed to **Google Gemini**. By using a specialized system prompt, the LLM evaluates the domain's age, naming conventions, and registration details to categorize the entity as "Legit," "Suspicious," or "High Risk."
- **Output Layer (Data Logging):** Final verdicts and analysis are structured and pushed directly into an **Airtable** database, creating a centralized log for future auditing.

Key Engineering Features

- **Resilient Design:** By bypassing front-end scraping in favor of standard protocols (RDAP) and API-driven search, the system is significantly more robust against website structure changes.
- **Structured Output:** The pipeline forces the AI to output valid JSON, allowing for seamless data mapping and integration with downstream database nodes.

- **Intelligent Routing:** The branching logic allows the system to intelligently switch between manual URL processing and automated discovery, reducing user effort.

Summary

This project demonstrates an end-to-end automation workflow that solves a common data-gathering challenge with a production-grade approach. It replaces manual, error-prone verification processes with an automated, AI-augmented pipeline that provides actionable intelligence.

Company Verification - Google Chrome

localhost:5678/form-test/ac228559-85e9-4efe-b63c-10e9ae...

This is a test version of your form

Company Verification

Company Name *

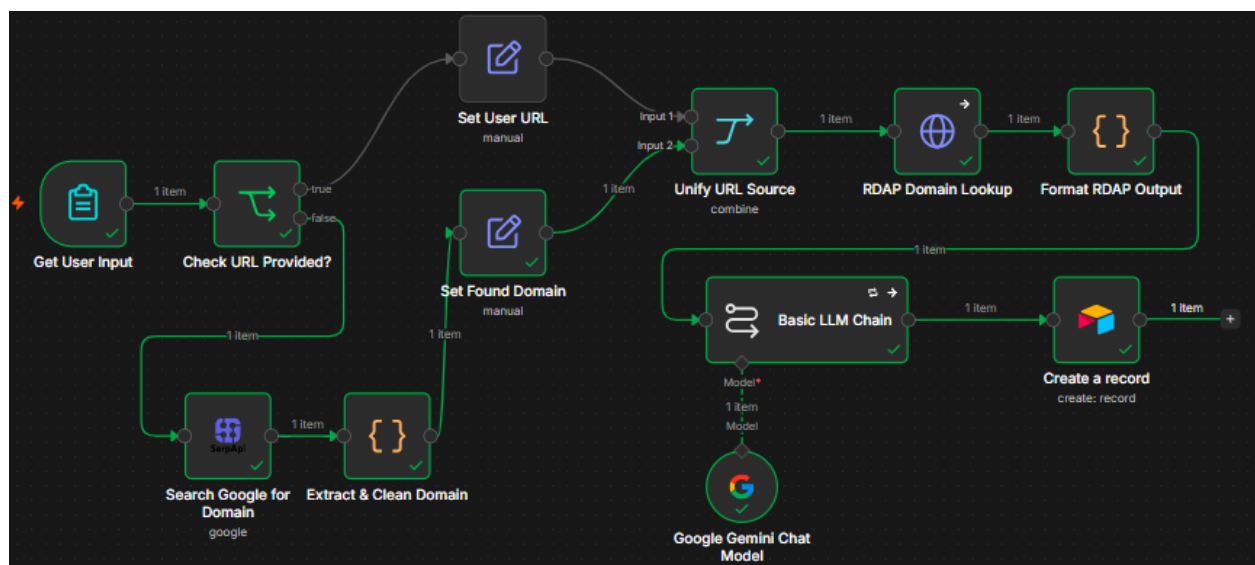
Google

Company URL

e.g. https://www.google.com

Submit

Form automated with n8n



Company Verification - Google Chrome

localhost:5678/form-test/ac228559-85e9-4efe-b63c-10e9ae9dc1a0

This is a test version of your form

Company Verification

Company Name *

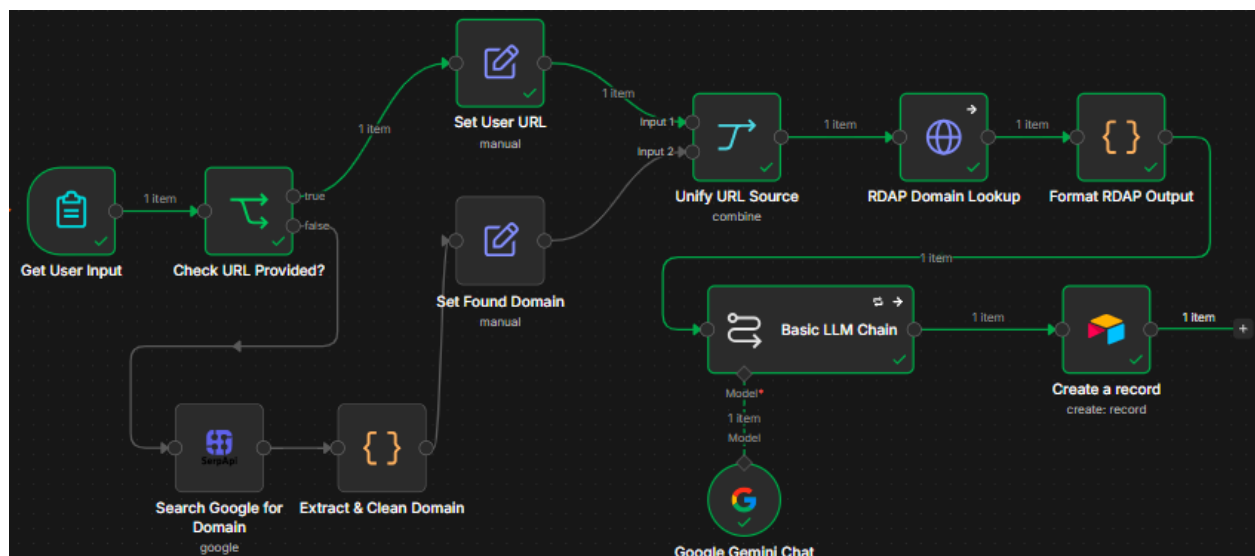
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Company URL

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Submit

Form automated with  n8n



Airtable:

JoAutomated Risk Assessment Pipeline

DataAutomationsInterfacesForms

Upgrade

Company

Grid view

Hide fieldsFilterGroupSortColorTeam

	Company Name	Created Date		Expiry Date		Registrar	Company URL
1	Google	8/6/2018	1:19pm	8/6/2027	1:19pm	MarkMonitor Inc.	about.google
2	DevelopersHub Corporatio...	11/20/2024	1:10pm	11/20/2026	1:10pm	Realtime Register B.V.	developershubcorp.com
3	Gamma	8/11/2020	7:26pm	8/11/2027	7:26pm	GoDaddy.com, LLC	gamma.app
+							

Hide fieldsFilterGroupSortColorTeam

#	Domain Age in days	Verdict	Analysis	Risk Score
	2,895.0	Legit	Mature domain with strong...	0.0
	598.0	Legit	The domain name matches...	0.0
	2,159.0	Legit	The domain perfectly matc...	0.0